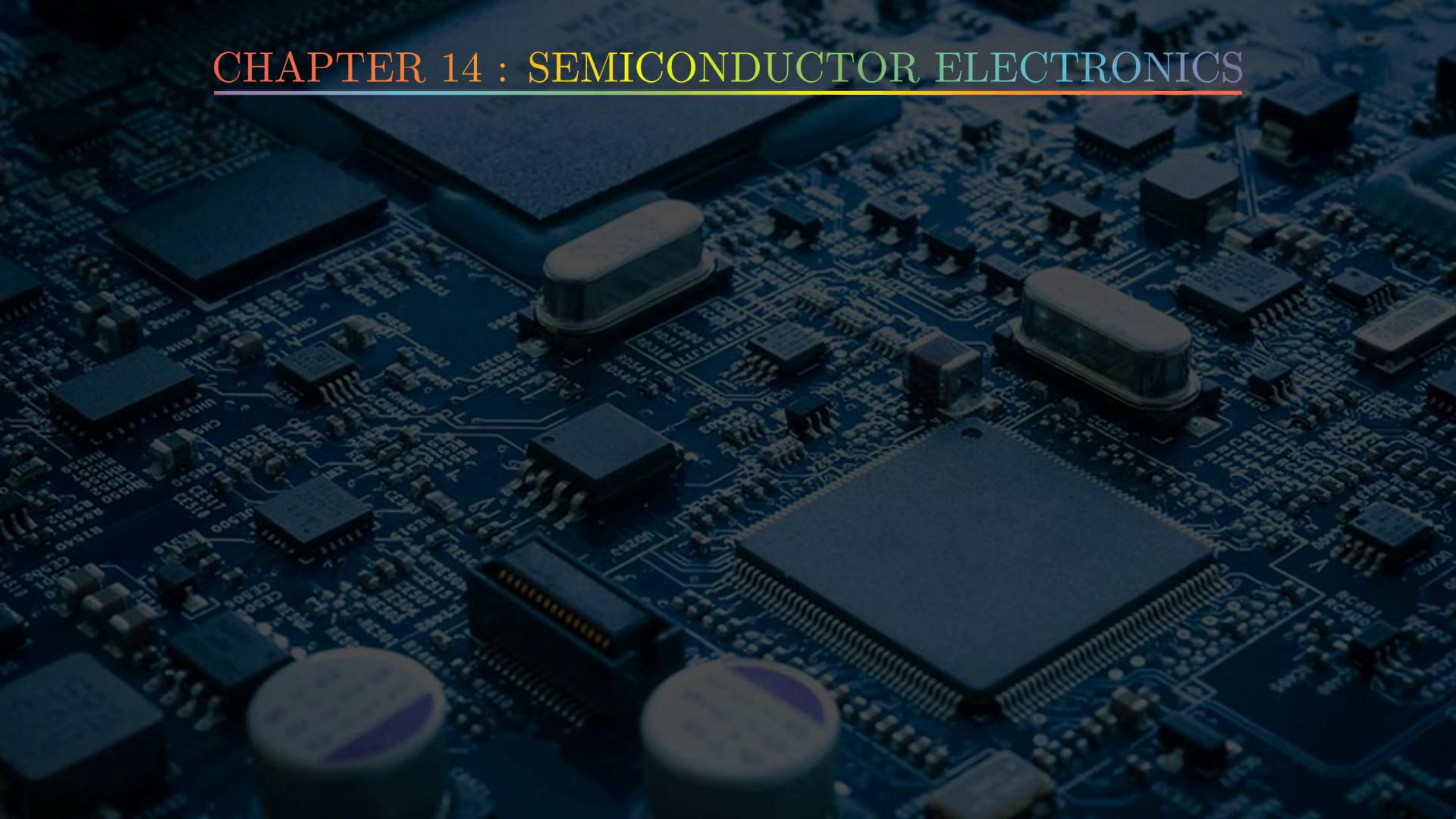


CHAPTER 14 : SEMICONDUCTOR ELECTRONICS

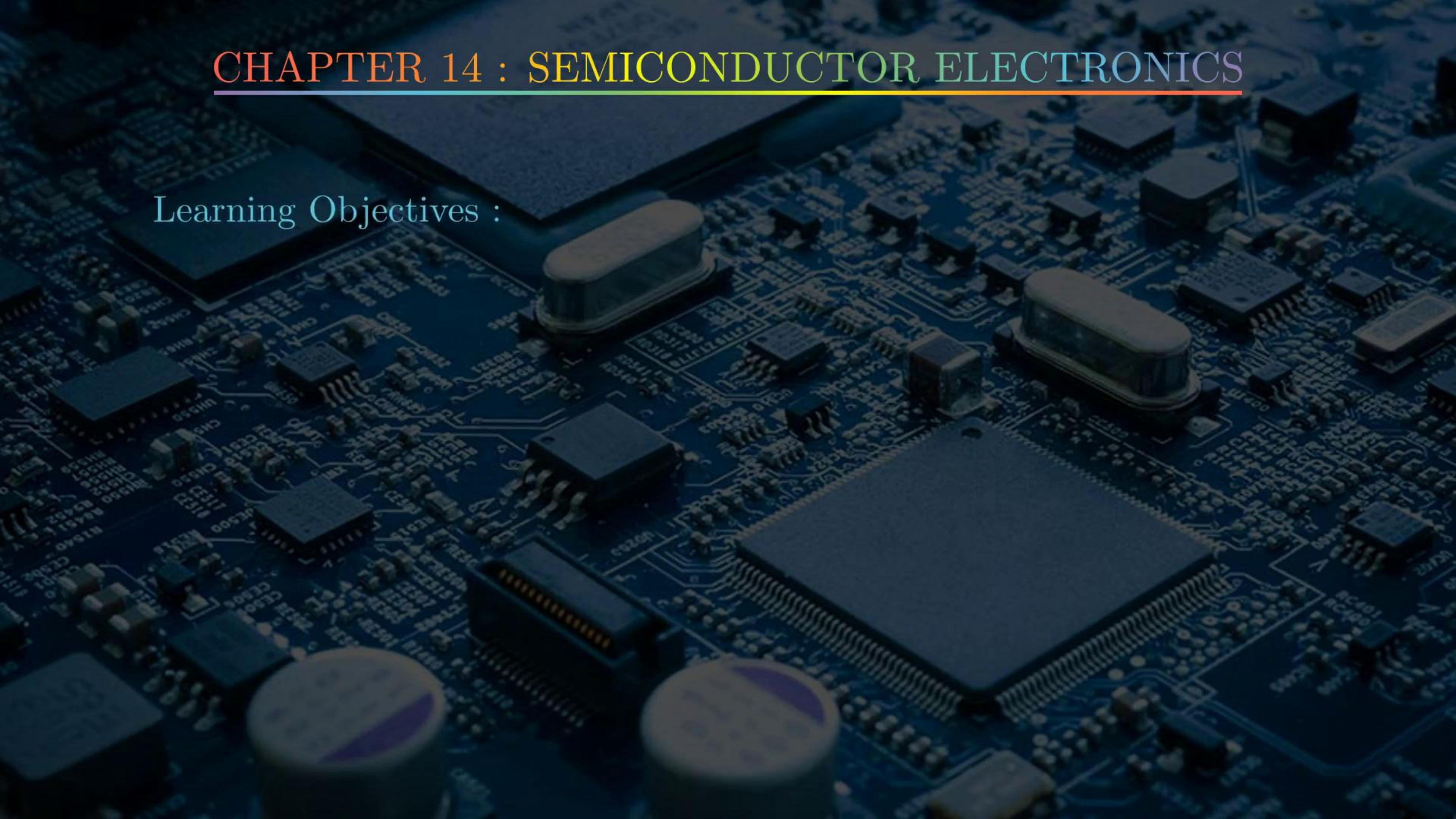


CHAPTER 14 : SEMICONDUCTOR ELECTRONICS

Learning Objectives :

CHAPTER 14 : SEMICONDUCTOR ELECTRONICS

Learning Objectives :



CHAPTER 14 : SEMICONDUCTOR ELECTRONICS

Learning Objectives :

- Energy bands in conductors, semiconductors and insulators (qualitative ideas only)

CHAPTER 14 : SEMICONDUCTOR ELECTRONICS

Learning Objectives :

- Energy bands in conductors, semiconductors and insulators (qualitative ideas only)
- Intrinsic and extrinsic semiconductors - p and n type

CHAPTER 14 : SEMICONDUCTOR ELECTRONICS

Learning Objectives :

- Energy bands in conductors, semiconductors and insulators (qualitative ideas only)
- Intrinsic and extrinsic semiconductors - p and n type
- p-n junction Semiconductor diode

CHAPTER 14 : SEMICONDUCTOR ELECTRONICS

Learning Objectives :

- Energy bands in conductors, semiconductors and insulators (qualitative ideas only)
- Intrinsic and extrinsic semiconductors - p and n type
- p-n junction Semiconductor diode
- I-V characteristics in forward and reverse bias

CHAPTER 14 : SEMICONDUCTOR ELECTRONICS

Learning Objectives :

- Energy bands in conductors, semiconductors and insulators (qualitative ideas only)
- Intrinsic and extrinsic semiconductors - p and n type
- p-n junction Semiconductor diode
- I-V characteristics in forward and reverse bias
- Application of junction diode - diode as a rectifier.

Classification of Metals, Insulator And Semiconductor

Semiconductor Electronics: Material Devices and Simple circuit

ON THE BASIS OF CONDUCTIVITY

(1) For metals:

$$S \sim 10^2 - 10^8 \Omega m$$

$$\sigma \sim 10^2 - 10^8 S/m$$

They have high conductivity.

(2) For Semiconductors:

$$S \sim 10^5 - 10^6 \Omega m$$

$$\sigma \sim 10^5 - 10^6 S/m$$

They have intermediate conductivity to metals and insulators.

(3) For insulators:

$$S \sim 10^{11} - 10^{19} \Omega m$$

$$\sigma \sim 10^{11} - 10^{19} S/m$$

They have low conductivity

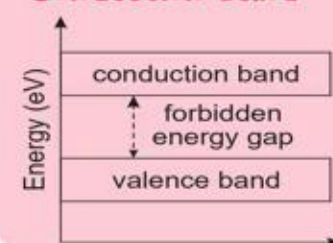
σ = electrical conductivity

ρ = resistivity

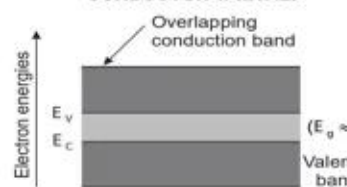
The band which is completely filled with electrons at OK is called valence band.

Conduction Band is completely empty at OK.

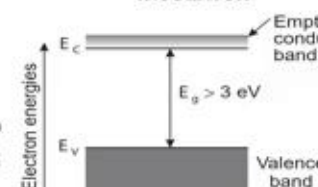
Energy band gap is the difference between valence band and conduction band



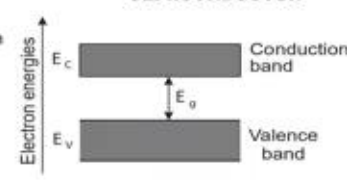
CONDUCTOR (METAL)



INSULATOR



SEMICONDUCTOR



SEMICONDUCTOR AND ITS TYPES

Semiconductors exhibit electrical conductivity between conductors and Non-Conductors.

INTRINSIC SEMICONDUCTORS

- (1) Pure Semiconductors are intrinsic Semiconductors.
- (2) $N_i = N_e = N_h$, where, N_e = No. of free electrons, N_h = No. of holes, N_i = intrinsic carrier Concentration
- (3) Examples:- Ge, Si

EXTRINSIC SEMICONDUCTORS

- (1) Impure or doped Semiconductors are said to be extrinsic Semiconductors
- (2) Impurities are added to improve Conductivity

N - type Semiconductor $N_e \gg N_h$

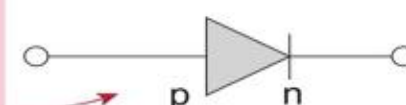
- (1) Electrons are majority charge carriers.
- (2) Holes are minority charge carriers.
- (3) Si or Ge doped with Pentavalent elements (P, As, Sb)

P - type Semiconductor $N_h \gg N_e$

- (1) Si or Ge doped with trivalent (B, Al) elements
- (2) Electrons are minority charge carriers.
- (3) Holes are majority charge carriers

THERMAL EQUILIBRIUM

The electron and hole Concentration in a Semiconductor in thermal equilibrium is given by, $n_e n_h = n_i^2$



P - N JUNCTION diode

P - N JUNCTION diode is the combination of P - type and N - type Semiconductor.
P - region has mobile majority holes and immobile - ve ions.
N - region has mobile majority free electrons and immobile positively charged ions.

POTENTIAL BARRIERS

Potential barrier is the potential difference developed across depletion region.

$$V_B = 0.7 \text{ for Silicon}$$

$$= 0.3 \text{ for germanium}$$

FORWARD BIAS

In Forward Bias

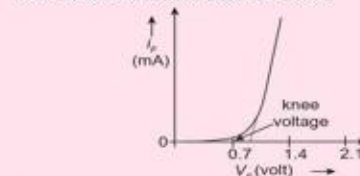
- (1) +ve terminal to P - Side
- (2) -ve terminal to N - Side
- (3) depletion layer reduced
- (4) diffusion current increases

REVERSE BIAS

In Reverse Bias

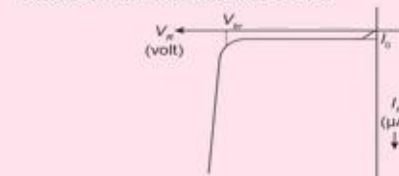
- (1) -ve terminal to P - Side
- (2) +ve terminal to N - Side
- (3) depletion layer increases
- (4) diffusion current increases

Forward characteristics curve



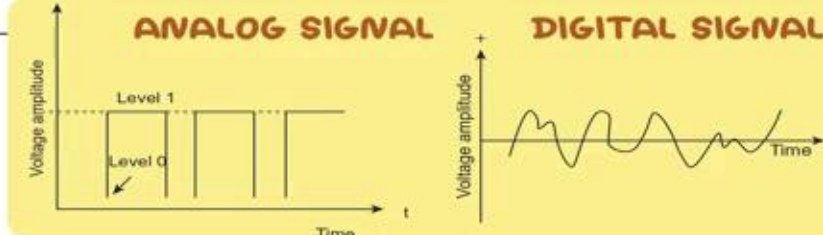
Knee or cut in voltage
Ge $\rightarrow$ 0.3 V, Si $\rightarrow$ 0.7 V

Reverse characteristic curve

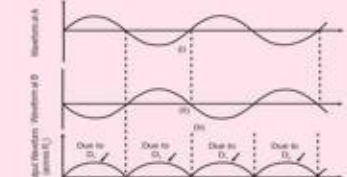
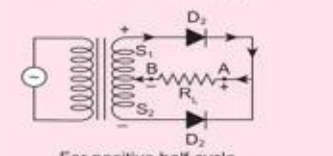
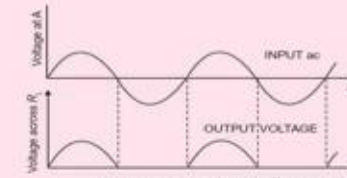
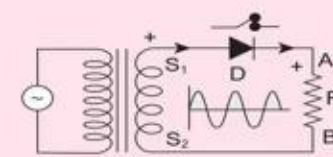


TRANSISTOR

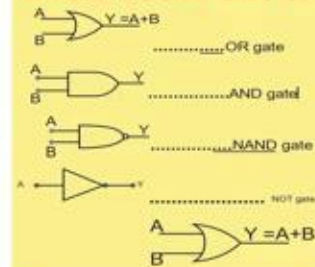
Transistor is a three terminal device
(1) Emitter (E)
(2) Base (B)
(3) Collector (C)



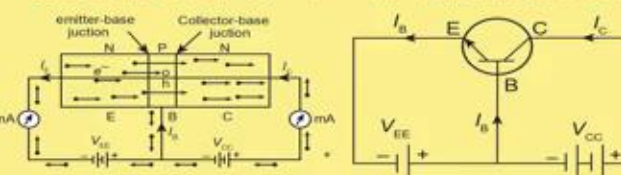
APPLICATIONS OF JUNCTION DIODE



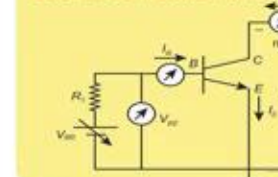
BASIC LOGIC GATES



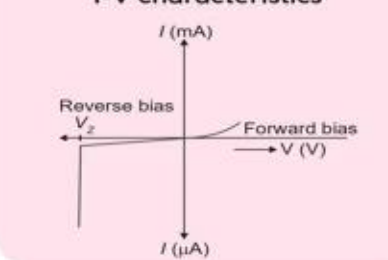
WORKING OF N - P - N TRANSISTOR



CE CONFIGURATION CHARACTERISTICS



SEMICONDUCTOR ZENER DIODES



ZENER BREAKDOWN

- THIS PHENOMENON TAKES PLACE IN
- (1) P - N JUNCTION HAVING 'high doping'
 - (2) P - N JUNCTION HAVING thin depletion layer
- Here, P - N JUNCTION does NOT damage permanently

AVALANCHE BREAKDOWN

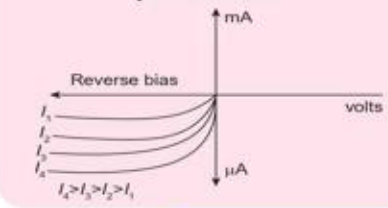
- THIS PHENOMENON TAKES PLACE IN
- (1) P - N JUNCTION HAVING 'low doping'
 - (2) P - N JUNCTION HAVING thick depletion layer.
- Here, P - N JUNCTION damages permanently due to abruptly increment of minorities during repetitive collisions.

PHOTODIODE

The symbol of photodiode is

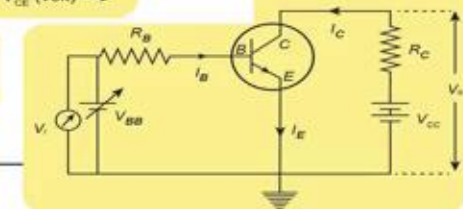


I-V characteristics of a photodiode

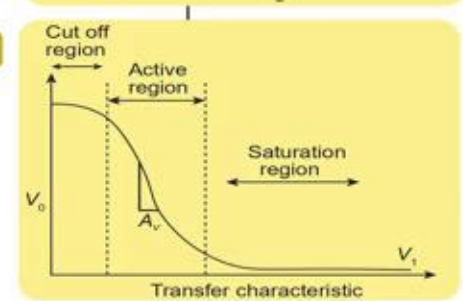
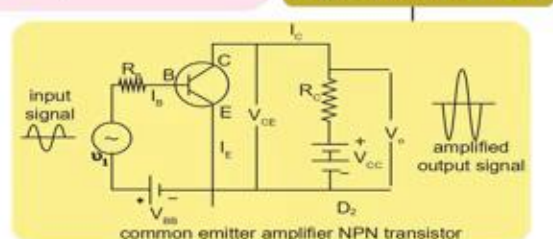


APPLICATION OF TRANSISTOR

AS A SWITCH

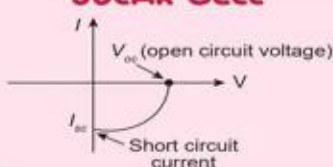


AS AN AMPLIFIER



I-V CHARACTERISTICS OF A SOLAR CELL.

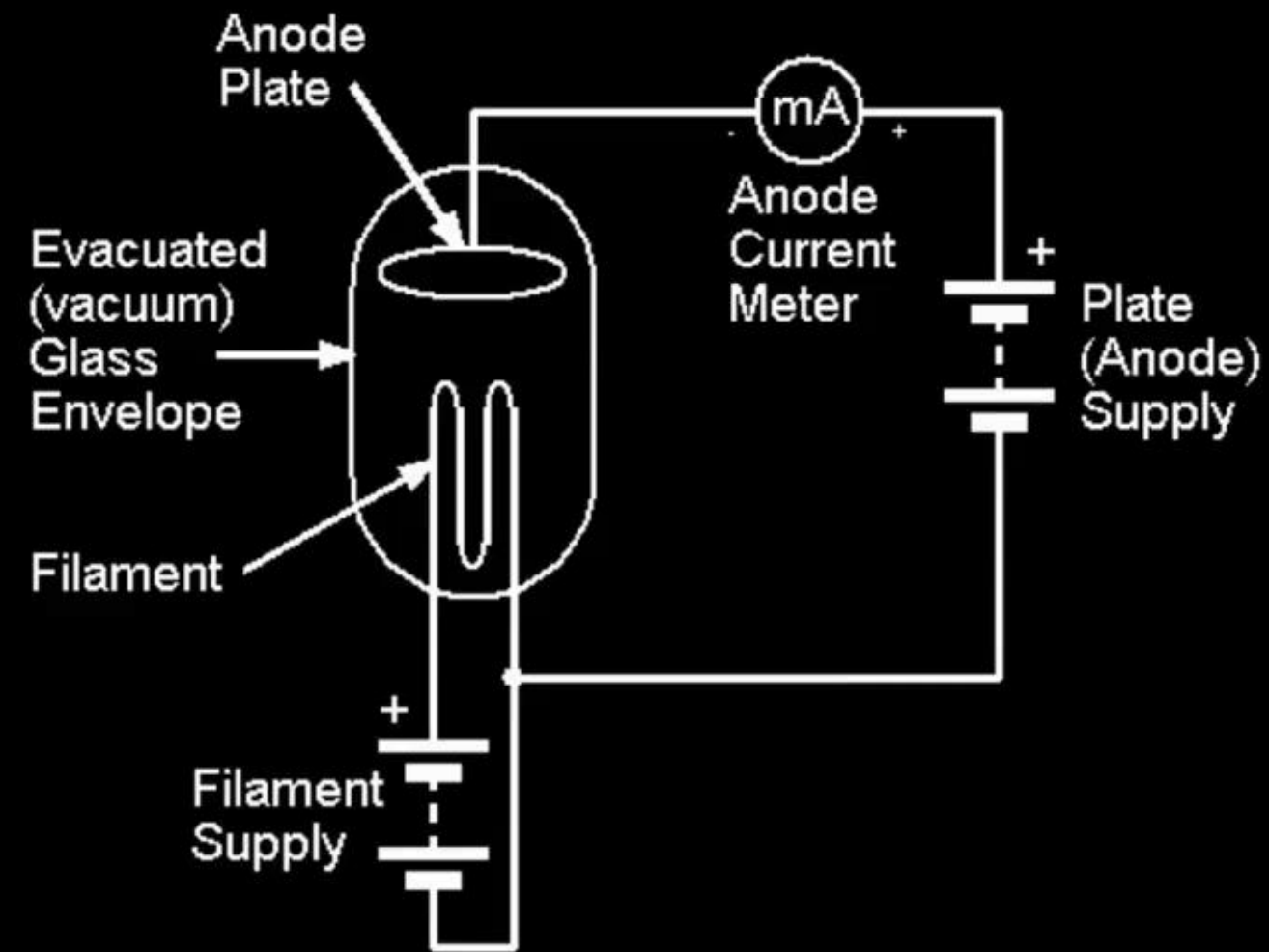
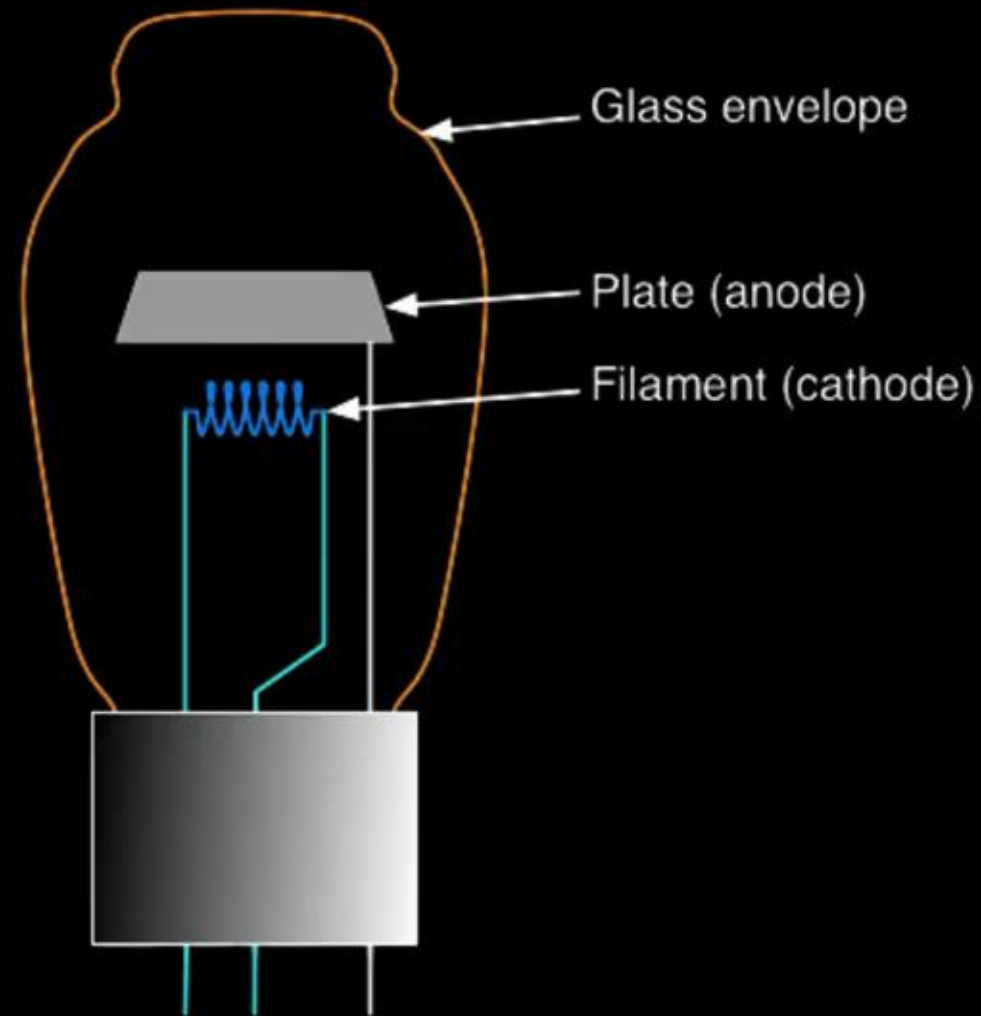
SOLAR CELL



Introduction

Introduction

- Before 1948, **vacuum tubes (valves)** like diode and triode were used as electronic devices.

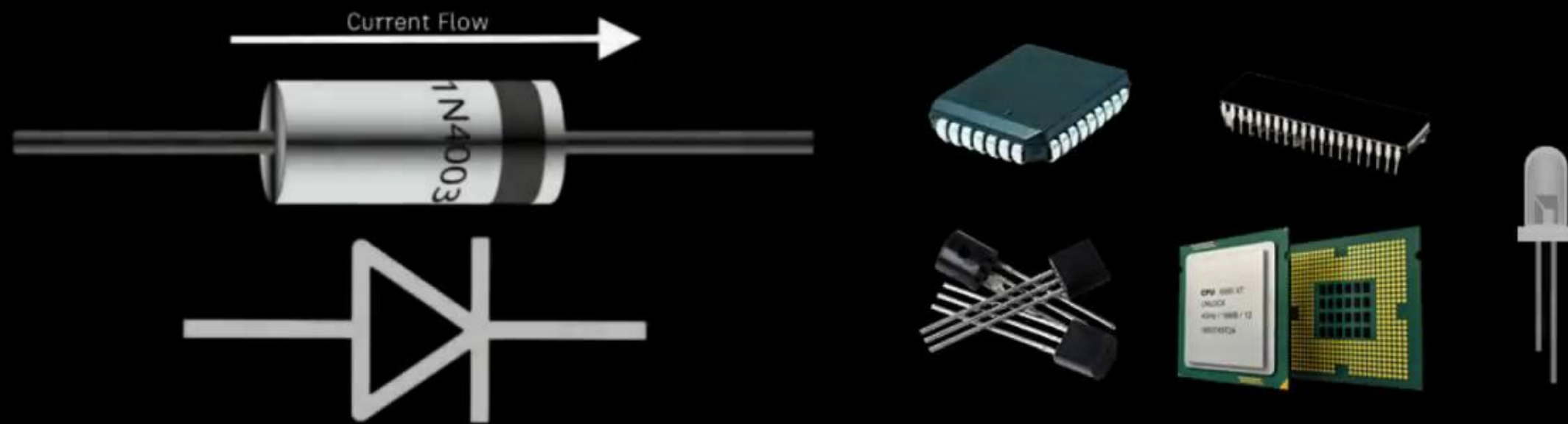


Introduction

- Before 1948, **vacuum tubes (valves)** like diode and triode were used as electronic devices.
- Vacuum tubes were **bulky, consumed high power, required high voltages, and had low reliability.**

Introduction

- Before 1948, **vacuum tubes (valves)** like diode and triode were used as electronic devices.
- Vacuum tubes were **bulky, consumed high power, required high voltages**, and had **low reliability**.
- **Semiconductors** (discovered in 1930s) allowed control of charge carriers inside solids without vacuum or heating.



Introduction

- Before 1948, **vacuum tubes (valves)** like diode and triode were used as electronic devices.
- Vacuum tubes were **bulky, consumed high power, required high voltages, and had low reliability.**
- **Semiconductors** (discovered in 1930s) allowed control of charge carriers inside solids without vacuum or heating.
- Advantages of semiconductors: **small size, low power, low voltage operation, long life, high reliability.**

Introduction

- Before 1948, **vacuum tubes (valves)** like diode and triode were used as electronic devices.
- Vacuum tubes were **bulky, consumed high power, required high voltages, and had low reliability.**
- **Semiconductors** (discovered in 1930s) allowed control of charge carriers inside solids without vacuum or heating.
- Advantages of semiconductors: **small size, low power, low voltage operation, long life, high reliability.**
- This chapter introduces **semiconductor physics, diodes, A few circuits illustrating their applications.**

Classification of Metals, Conductors and Semiconductors

Classification of Metals, Conductors and Semiconductors

- **Metals:** Very low resistivity (10^{-2} - $10^{-8} \Omega m$) and high conductivity (10^2 - $10^8 S m^{-1}$).

Classification of Metals, Conductors and Semiconductors

- **Metals:** Very low resistivity (10^{-2} - $10^{-8} \Omega m$) and high conductivity (10^2 - $10^8 S m^{-1}$).
- **Insulators:** Very high resistivity (10^{11} - $10^{19} \Omega m$) and very low conductivity.

Classification of Metals, Conductors and Semiconductors

- **Metals:** Very low resistivity (10^{-2} - $10^{-8} \Omega m$) and high conductivity (10^2 - $10^8 S m^{-1}$).
- **Insulators:** Very high resistivity (10^{11} - $10^{19} \Omega m$) and very low conductivity.
- **Semiconductors:** They have resistivity or conductivity intermediate to metals and insulators. (10^{-5} - $10^6 \Omega m$). Conductivity varies with **temperature, light, doping**.

Classification of Metals, Conductors and Semiconductors

- **Metals:** Very low resistivity (10^{-2} - $10^{-8} \Omega m$) and high conductivity (10^2 - $10^8 S m^{-1}$).
- **Insulators:** Very high resistivity (10^{11} - $10^{19} \Omega m$) and very low conductivity.
- **Semiconductors:** They have resistivity or conductivity intermediate to metals and insulators. (10^{-5} - $10^6 \Omega m$). Conductivity varies with **temperature, light, doping**.
- **Types of semiconductors:** **Elemental (Si, Ge), Compound** - Inorganic (CdS, GaAs, CdSe, InP, etc.), Organic (anthracene), and Polymers (polyaniline, polypyrrole).

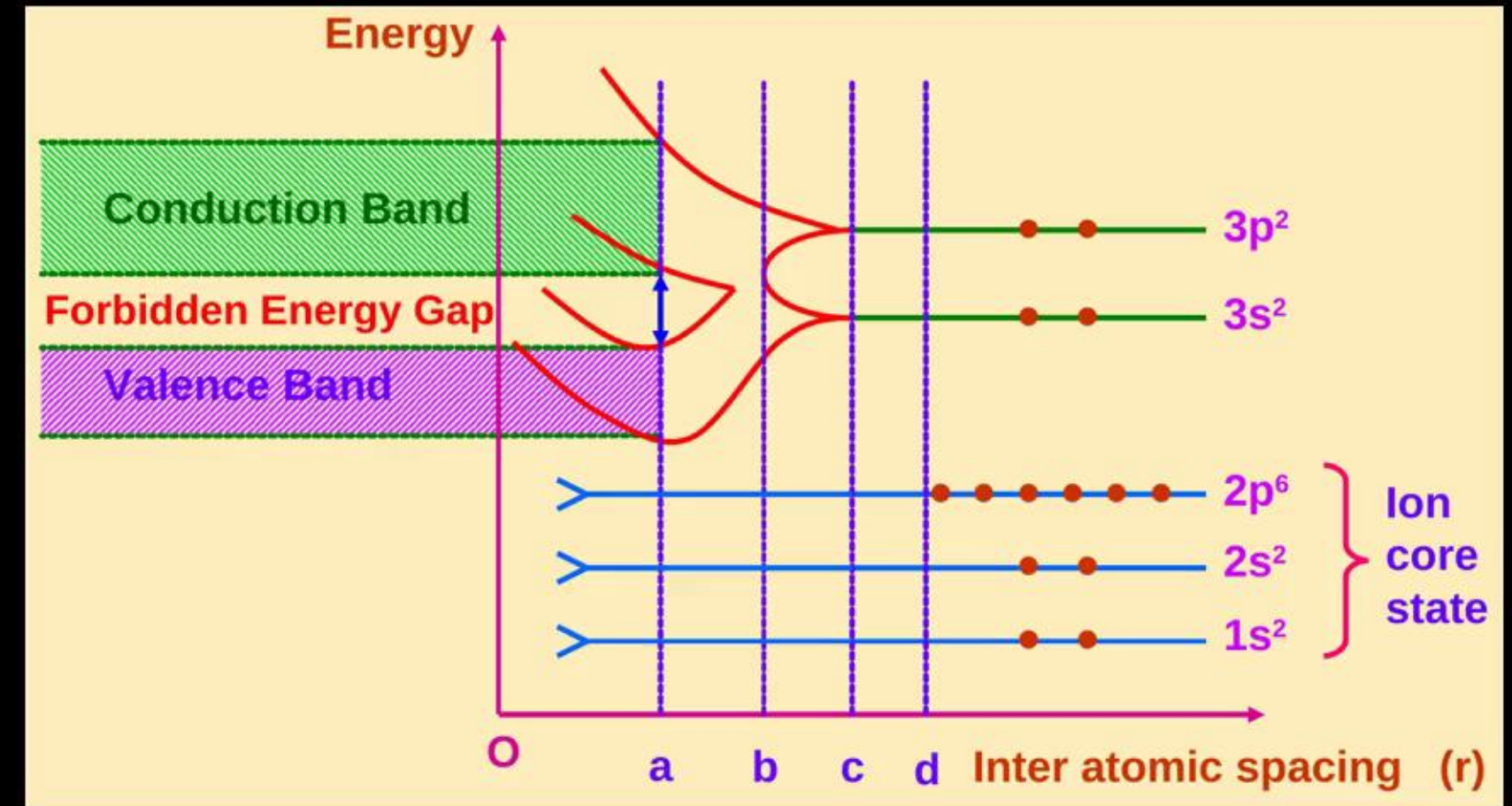
Classification of Metals, Conductors and Semiconductors

- **Metals:** Very low resistivity (10^{-2} - $10^{-8} \Omega m$) and high conductivity (10^2 - $10^8 S m^{-1}$).
- **Insulators:** Very high resistivity (10^{11} - $10^{19} \Omega m$) and very low conductivity.
- **Semiconductors:** They have resistivity or conductivity intermediate to metals and insulators. (10^{-5} - $10^6 \Omega m$). Conductivity varies with **temperature, light, doping**.
- Types of semiconductors: **Elemental (Si, Ge), Compound** - Inorganic (CdS, GaAs, CdSe, InP, etc.), Organic (anthracene), and Polymers (polyaniline, polypyrrole).
- Order of conductivity: **Metals $\gg$ Semiconductors $\gg$ Insulators.**

Energy Band Formation

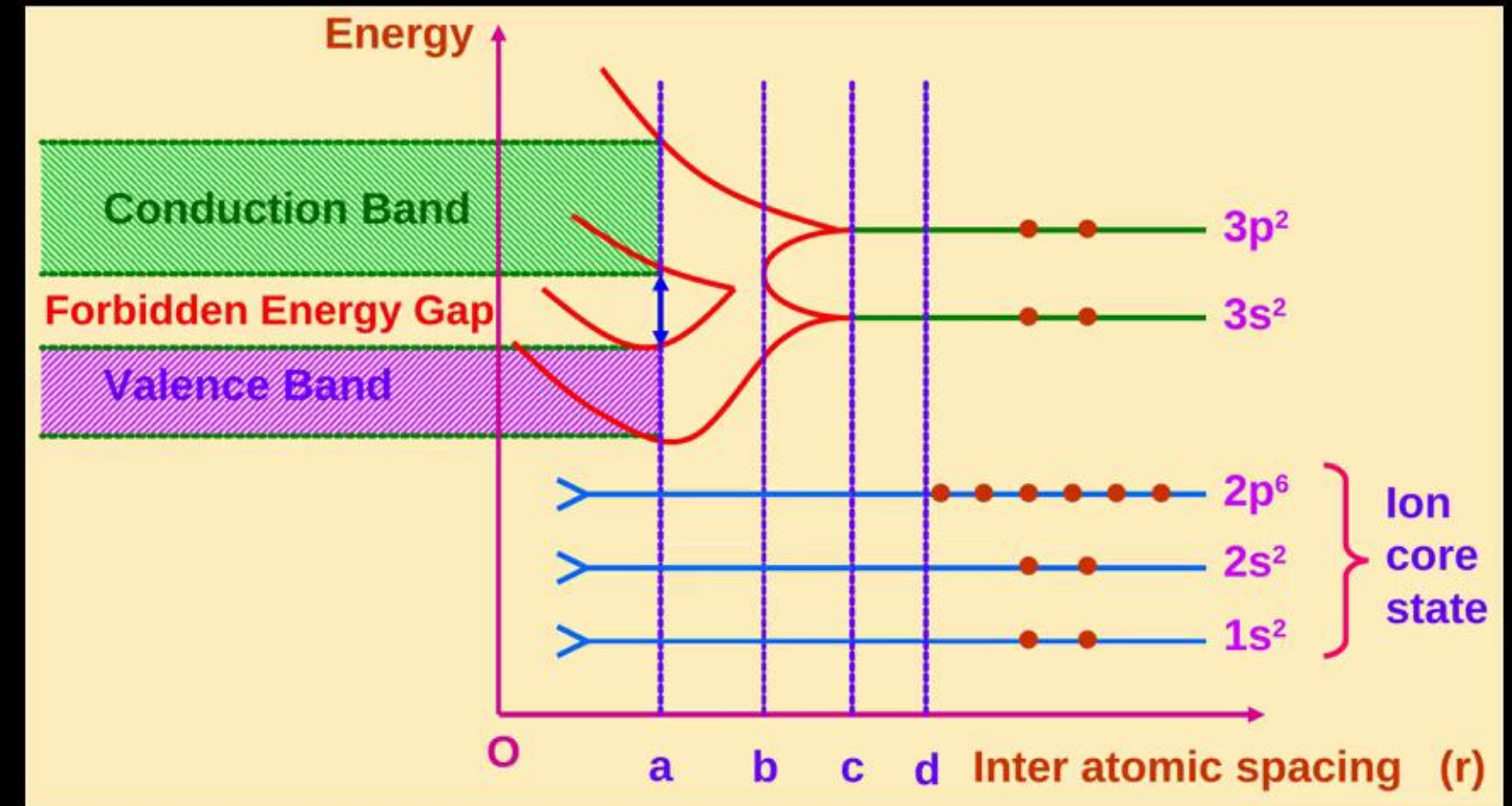
Energy Band Formation

- In an **isolated atom**, electrons occupy discrete energy levels.



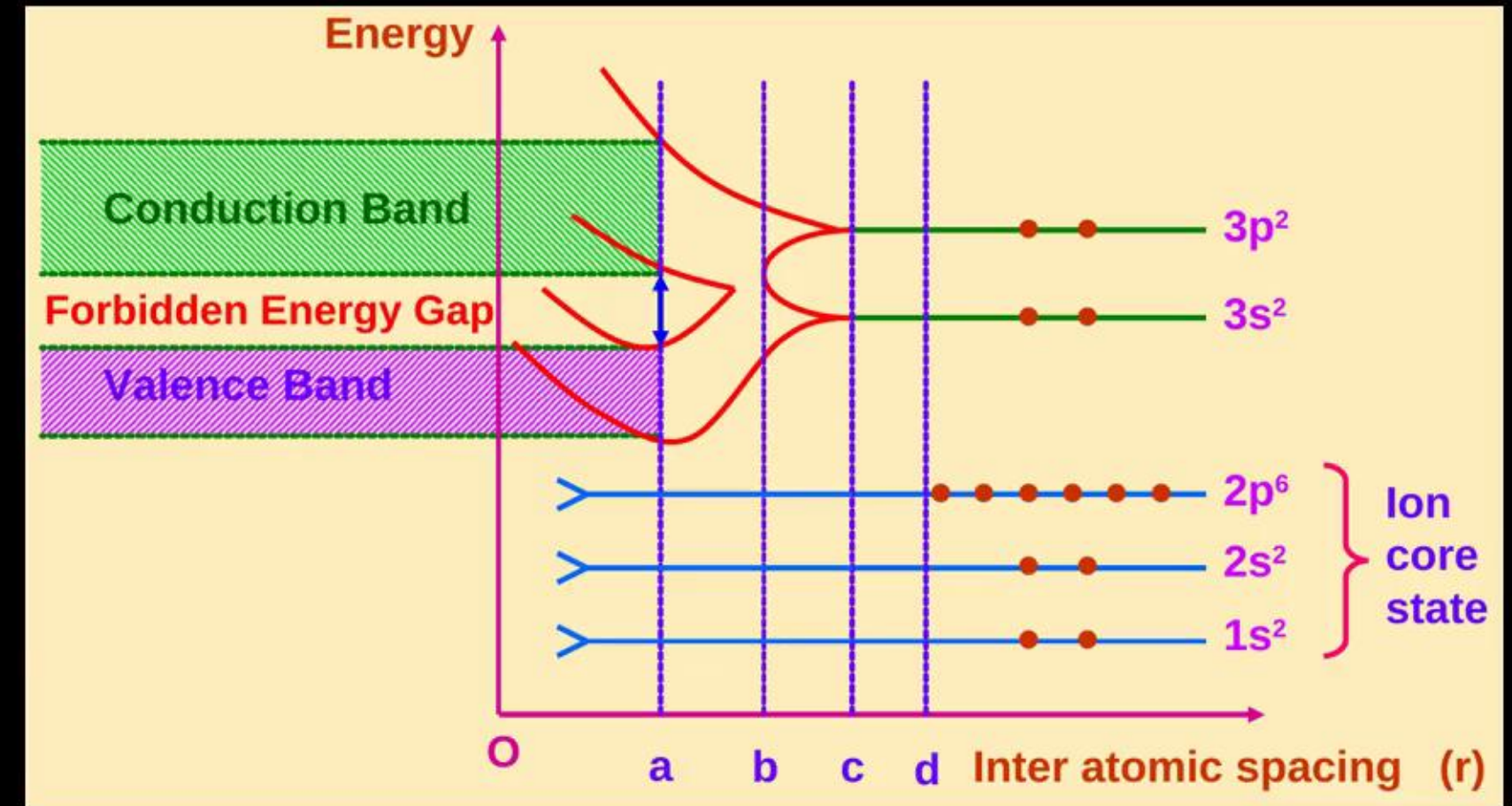
Energy Band Formation

- In an **isolated atom**, electrons occupy discrete energy levels.
- In a solid, atoms come close $\Rightarrow$ outer orbitals overlap.



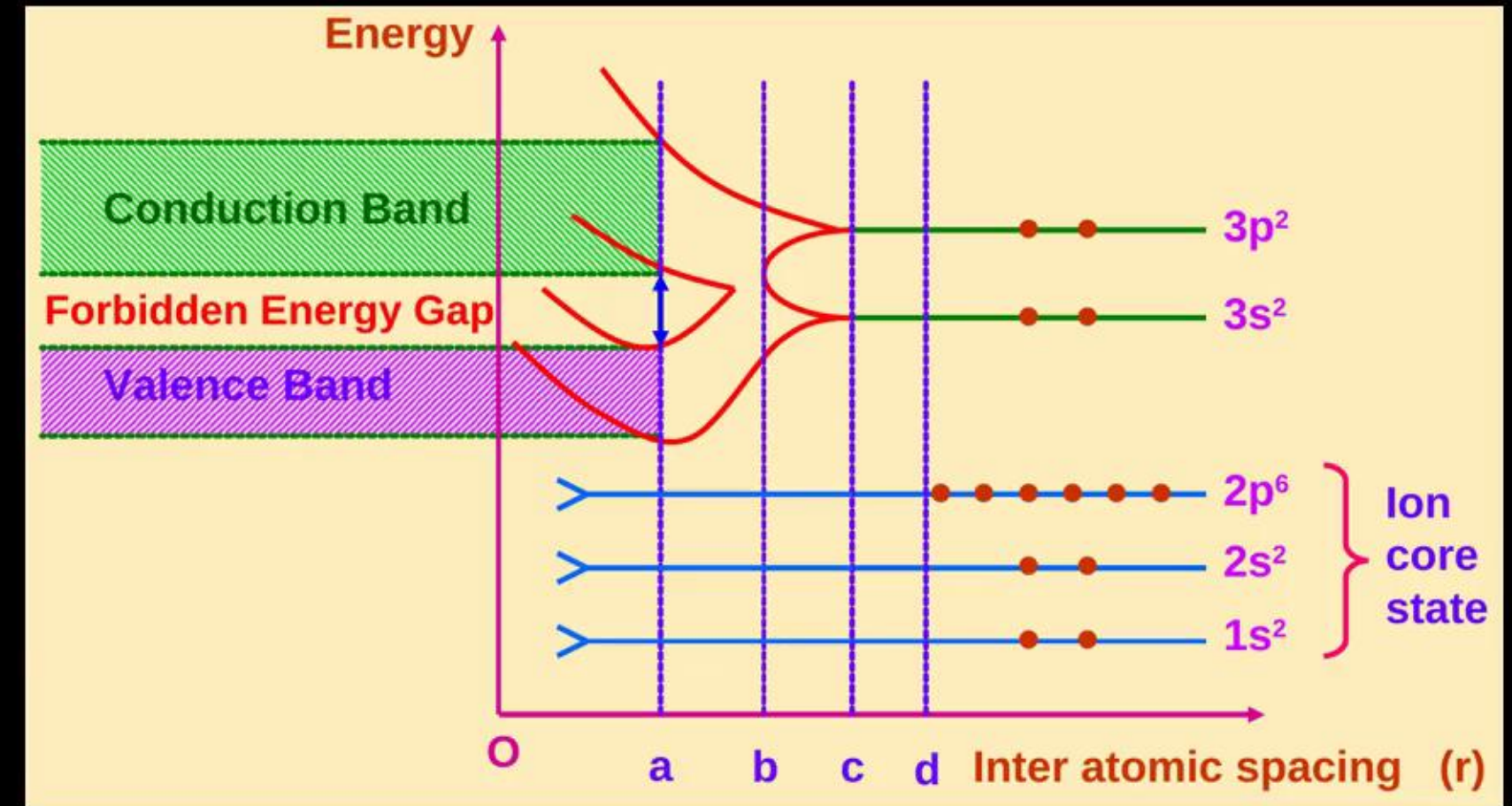
Energy Band Formation

- In an **isolated atom**, electrons occupy discrete energy levels.
- In a solid, atoms come close $\Rightarrow$ outer orbitals overlap.
- Discrete levels split into many closely spaced levels forming **energy bands**.



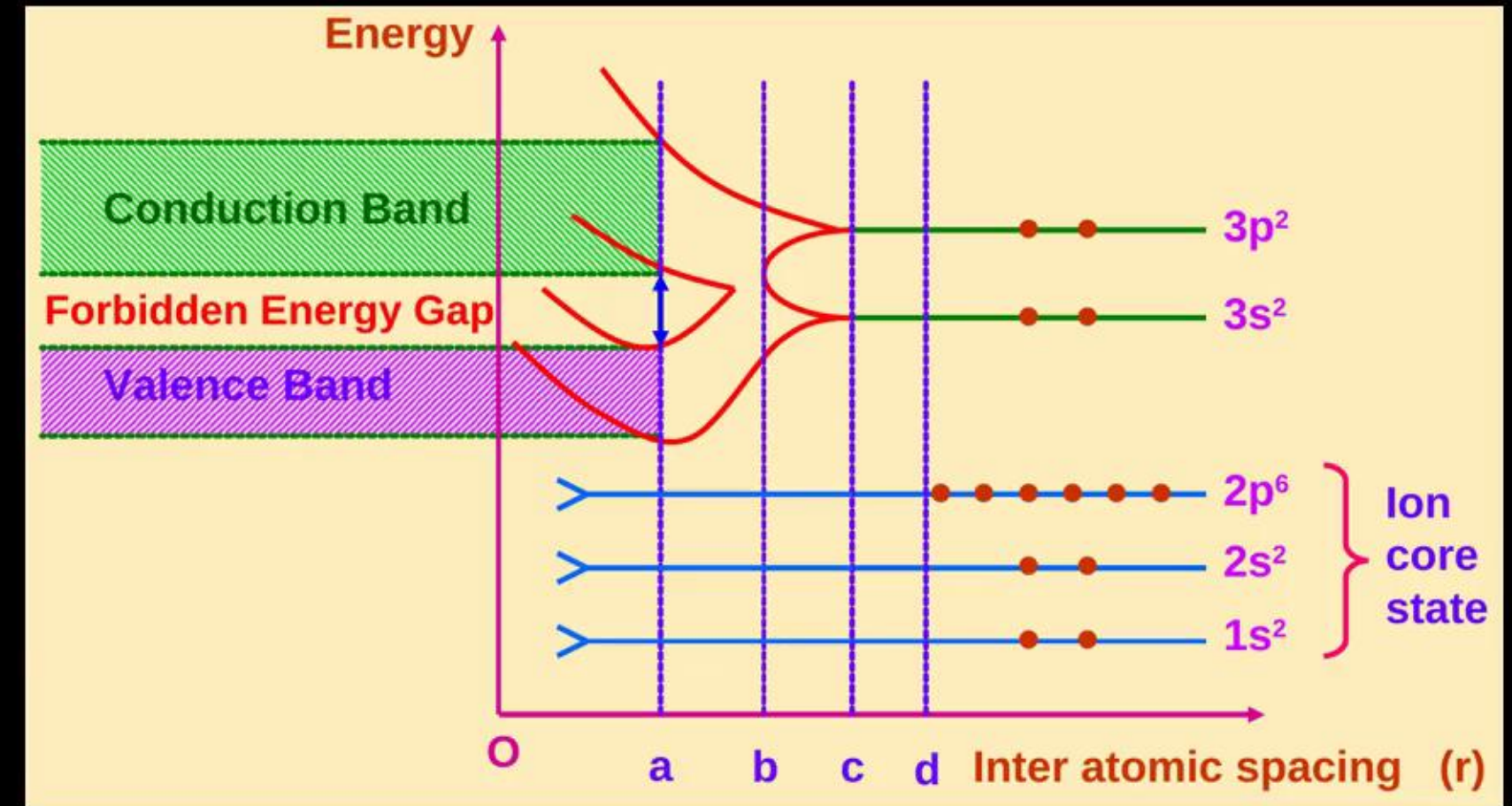
Energy Band Formation

- In an **isolated atom**, electrons occupy discrete energy levels.
- In a solid, atoms come close $\Rightarrow$ outer orbitals overlap.
- Discrete levels split into many closely spaced levels forming **energy bands**.
- Core electrons remain unaffected, only outer (valence) electrons form bands.



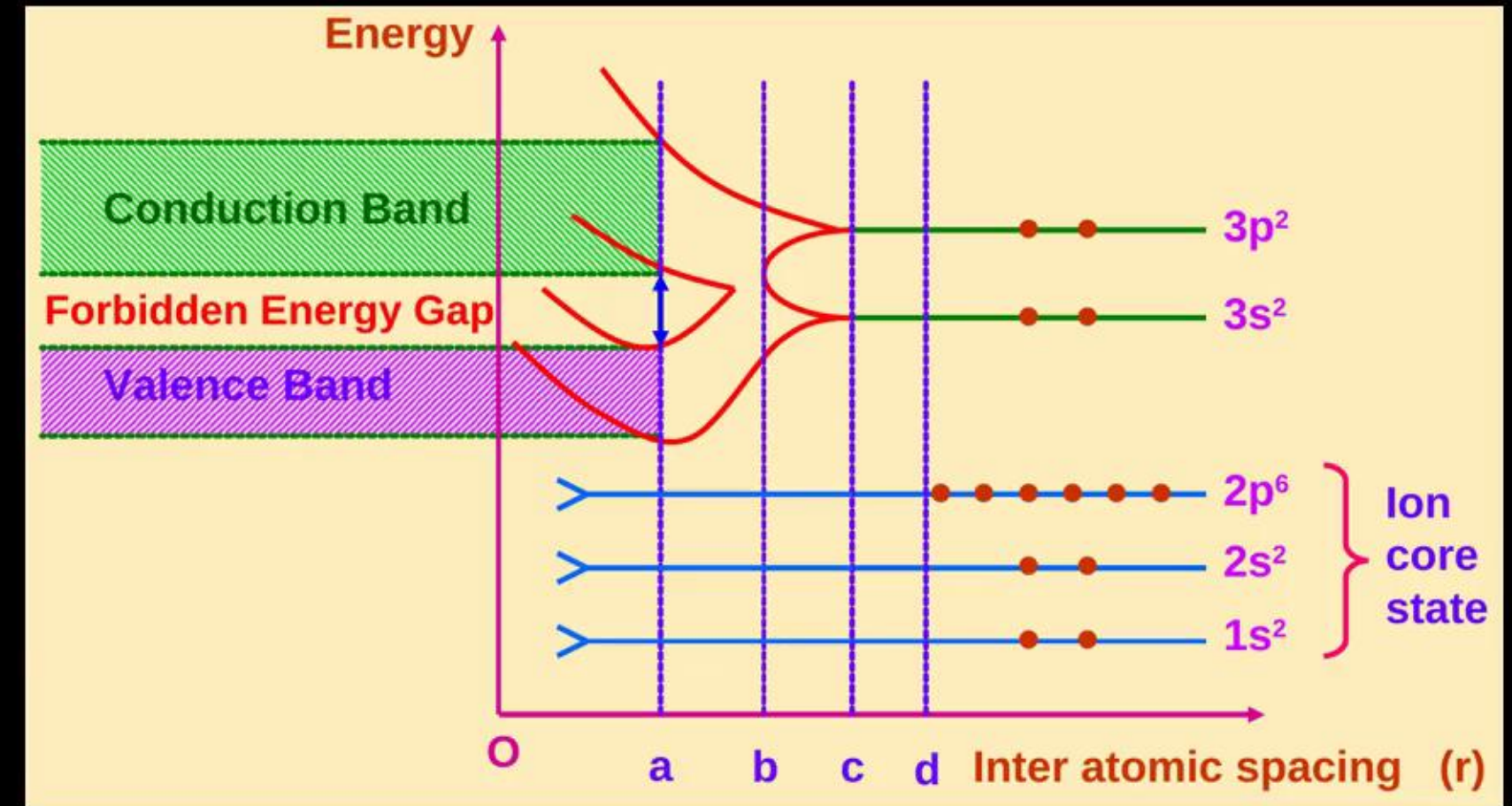
Energy Band Formation

- In an **isolated atom**, electrons occupy discrete energy levels.
- In a solid, atoms come close $\Rightarrow$ outer orbitals overlap.
- Discrete levels split into many closely spaced levels forming **energy bands**.
- Core electrons remain unaffected, only outer (valence) electrons form bands.
- Consider Si having Electronic configuration $1s^2, 2s^2, 2p^6, 3s^2, 3p^2$. (Atomic No. is 14)



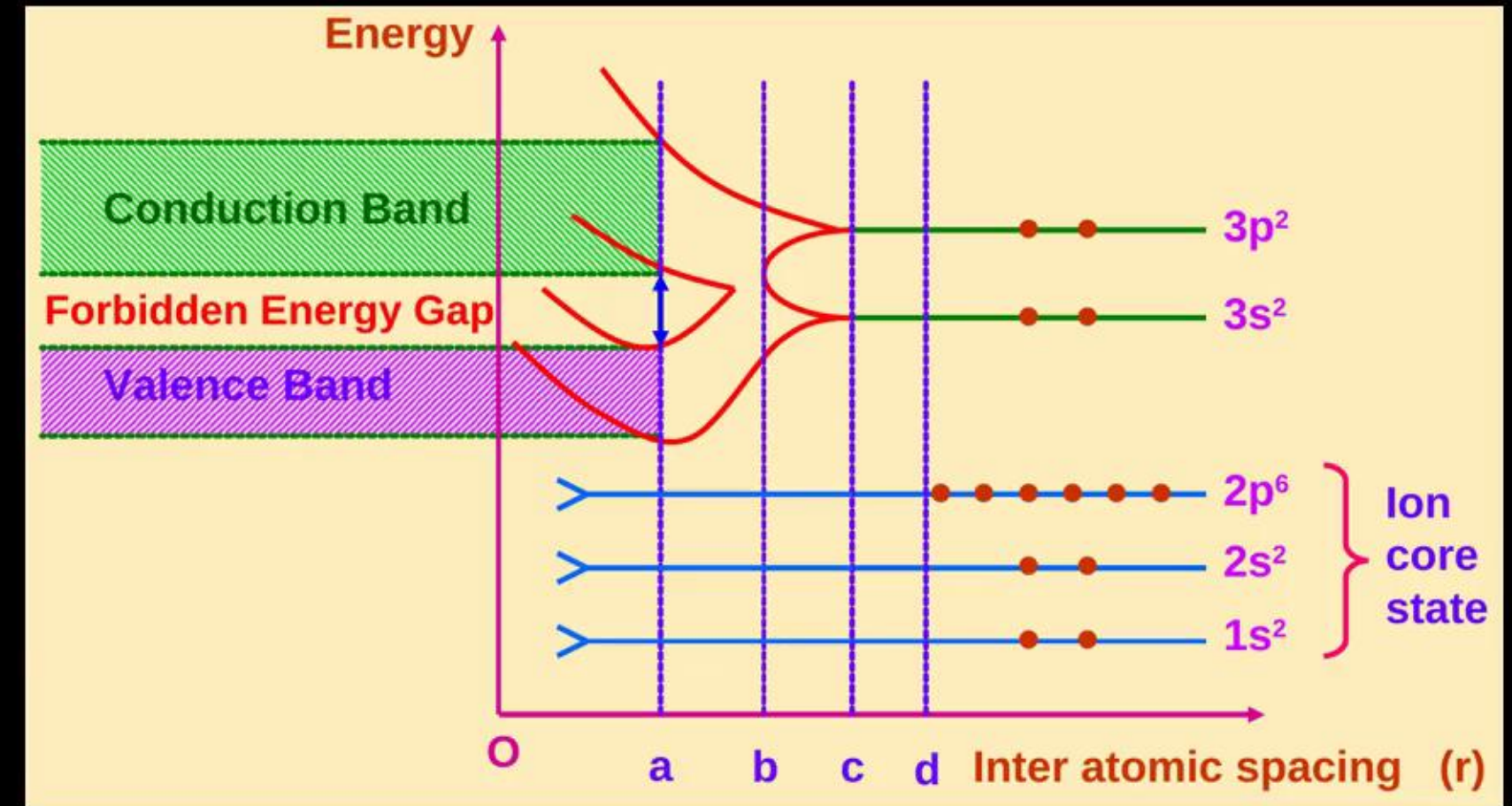
Energy Band Formation

- In an **isolated atom**, electrons occupy discrete energy levels.
- In a solid, atoms come close $\Rightarrow$ outer orbitals overlap.
- Discrete levels split into many closely spaced levels forming **energy bands**.
- Core electrons remain unaffected, only outer (valence) electrons form bands.
- Consider Si having Electronic configuration $1s^2, 2s^2, 2p^6, 3s^2, 3p^2$. (Atomic No. is 14)
- In Si/Ge crystal having N atoms.: $4N$ valence electrons $\Rightarrow 8N$ states split into 2 bands separated by **energy gap** E_g .



Energy Band Formation

- In an **isolated atom**, electrons occupy discrete energy levels.
- In a solid, atoms come close $\Rightarrow$ outer orbitals overlap.
- Discrete levels split into many closely spaced levels forming **energy bands**.
- Core electrons remain unaffected, only outer (valence) electrons form bands.
- Consider Si having Electronic configuration $1s^2, 2s^2, 2p^6, 3s^2, 3p^2$. (Atomic No. is 14)
- In Si/Ge crystal having N atoms.: $4N$ valence electrons $\Rightarrow 8N$ states split into 2 bands separated by **energy gap** E_g .



Valence Band, Conduction Band & Band Gap

Valence Band, Conduction Band & Band Gap

- **Valence Band (VB):** The energy band which includes the energy levels of the valence electrons (**bounded electrons**) is called the valence band. **It is completely filled at 0 K.**

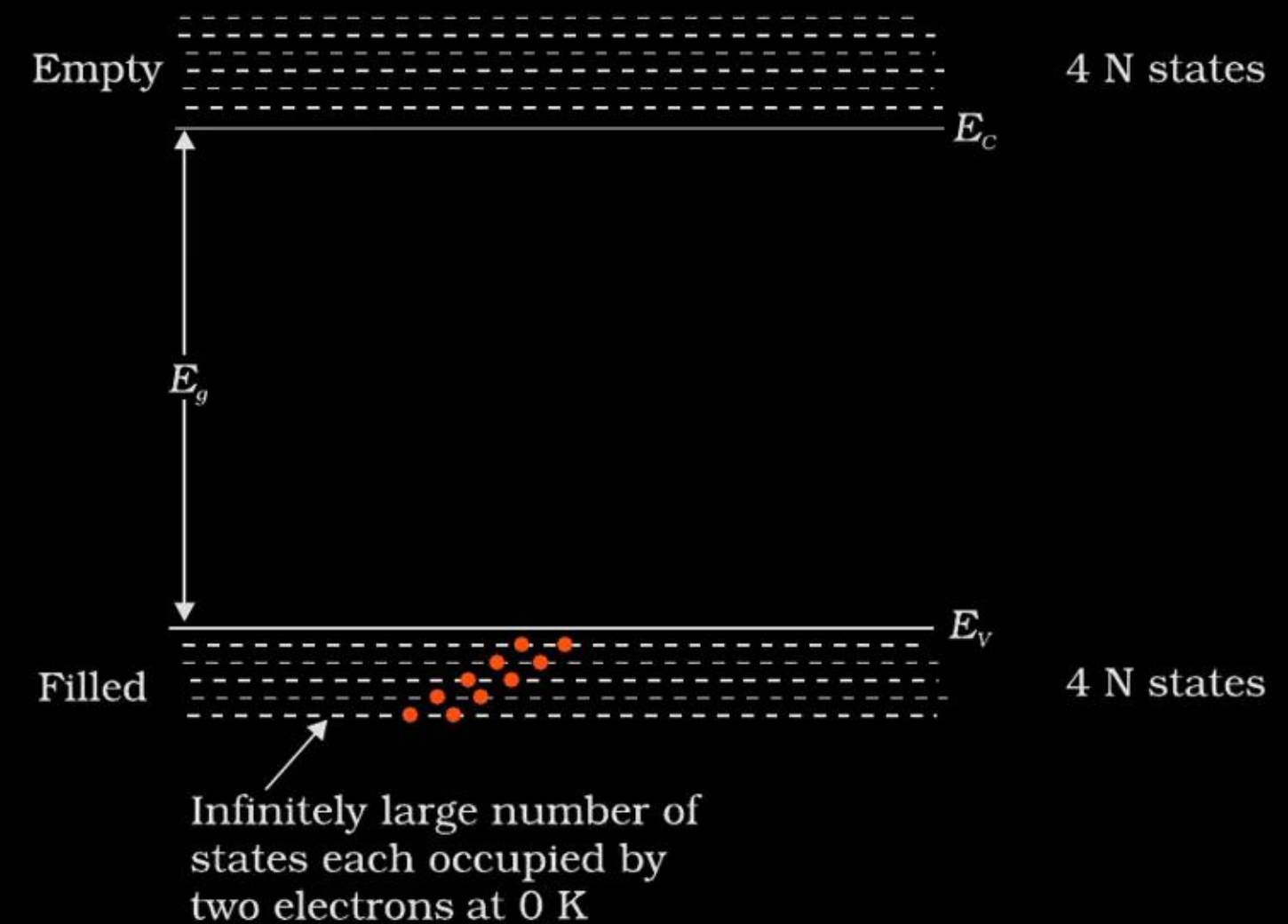


FIGURE 14.1 The energy band positions in a semiconductor at 0 K. The upper band, called the conduction band, consists of infinitely large number of closely spaced energy states. The lower band, called the valence band, consists of closely spaced completely filled energy states.

Valence Band, Conduction Band & Band Gap

- **Valence Band (VB):** The energy band which includes the energy levels of the valence electrons (**bounded electrons**) is called the valence band. **It is completely filled at 0 K.**
- **Conduction Band (CB):** Band above VB which contains free electrons; **electrons here are free to conduct** is called conduction band. It is normally empty at 0 K.

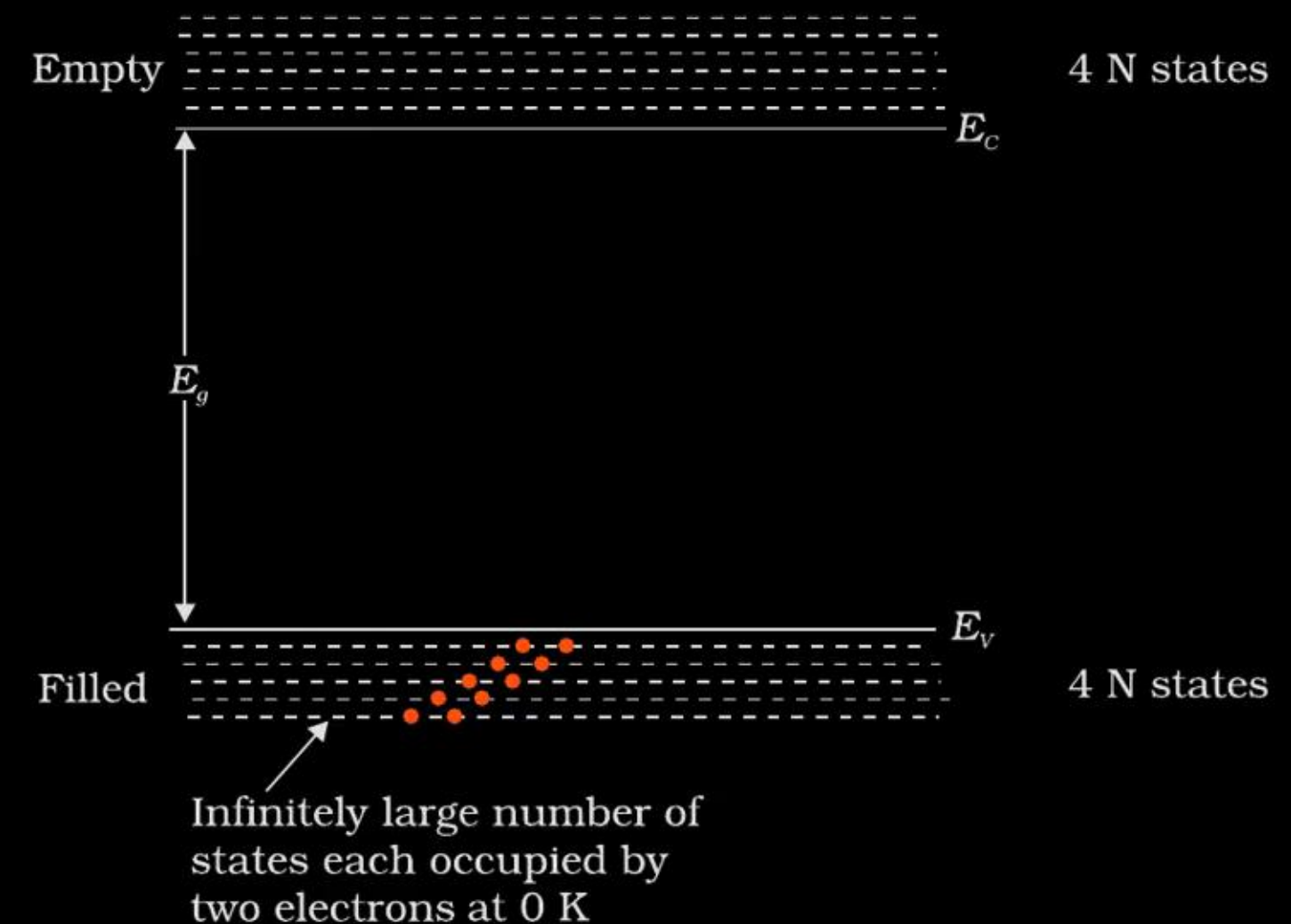


FIGURE 14.1 The energy band positions in a semiconductor at 0 K. The upper band, called the conduction band, consists of infinitely large number of closely spaced energy states. The lower band, called the valence band, consists of closely spaced completely filled energy states.

Valence Band, Conduction Band & Band Gap

- **Valence Band (VB):** The energy band which includes the energy levels of the valence electrons (**bounded electrons**) is called the valence band. **It is completely filled at 0 K.**
- **Conduction Band (CB):** Band above VB which contains free electrons; **electrons here are free to conduct** is called conduction band. It is normally empty at 0 K.
- **Energy Gap E_g :** Difference between top of VB (E_V) and bottom of CB (E_C).

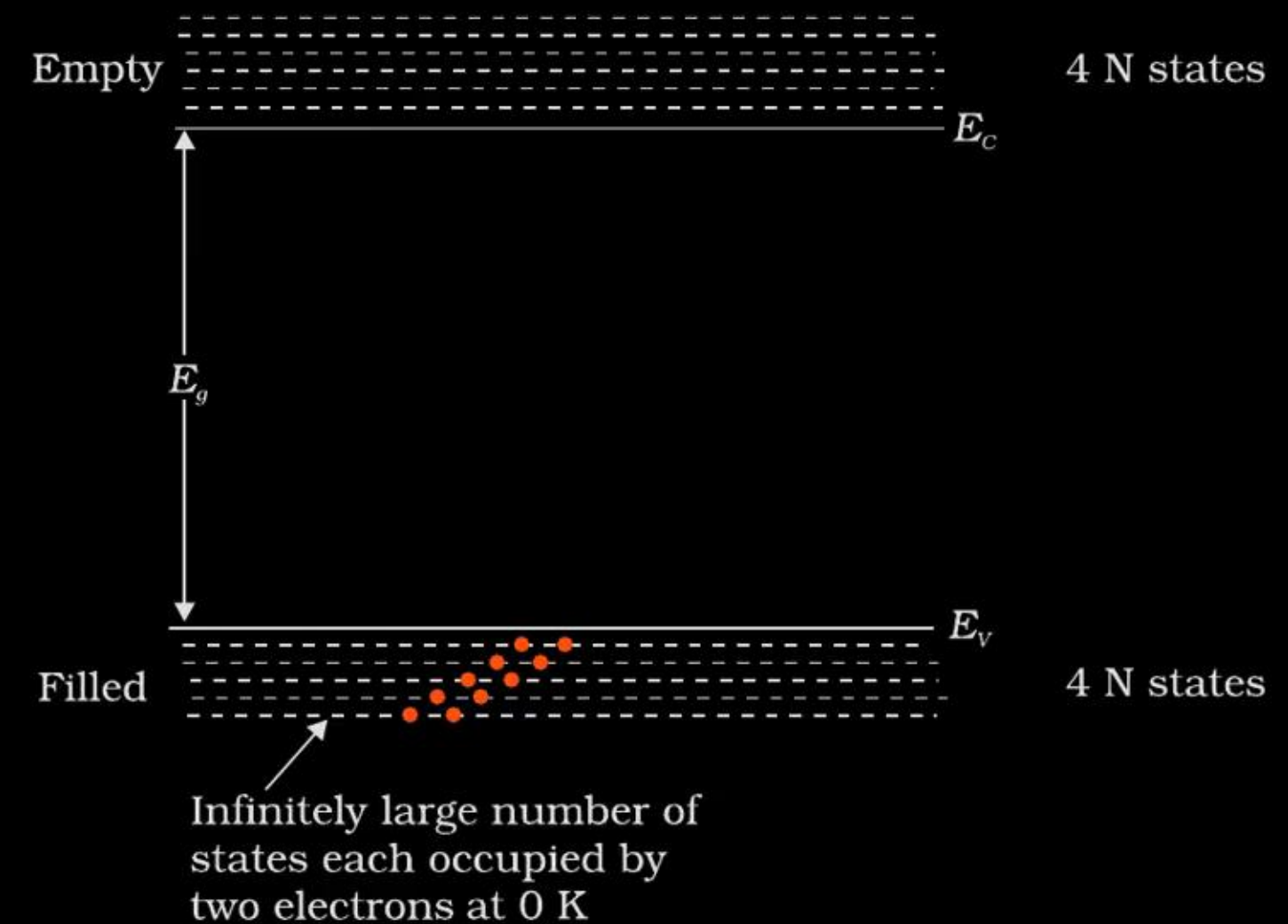


FIGURE 14.1 The energy band positions in a semiconductor at 0 K. The upper band, called the conduction band, consists of infinitely large number of closely spaced energy states. The lower band, called the valence band, consists of closely spaced completely filled energy states.

Valence Band, Conduction Band & Band Gap

- **Valence Band (VB):** The energy band which includes the energy levels of the valence electrons (**bounded electrons**) is called the valence band. **It is completely filled at 0 K.**
- **Conduction Band (CB):** Band above VB which contains free electrons; **electrons here are free to conduct** is called conduction band. It is normally empty at 0 K.
- **Energy Gap E_g :** Difference between top of VB (E_V) and bottom of CB (E_C).

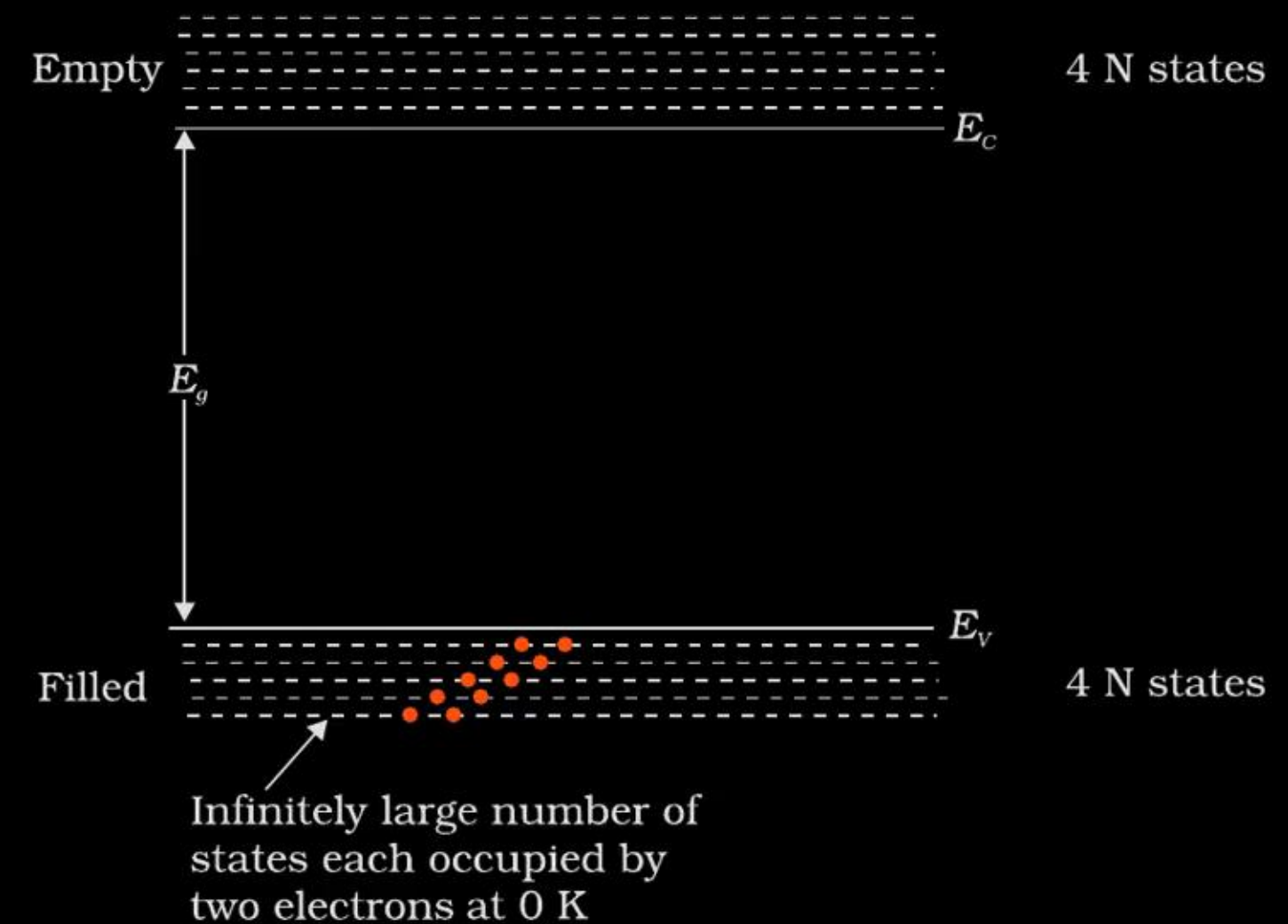
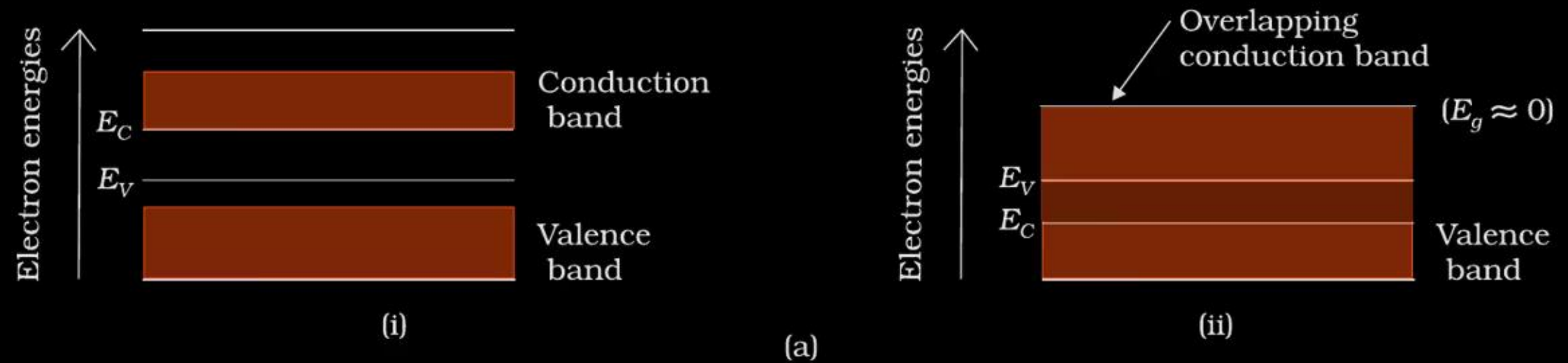


FIGURE 14.1 The energy band positions in a semiconductor at 0 K. The upper band, called the conduction band, consists of infinitely large number of closely spaced energy states. The lower band, called the valence band, consists of closely spaced completely filled energy states.

Classification on the Basis of Energy Bands

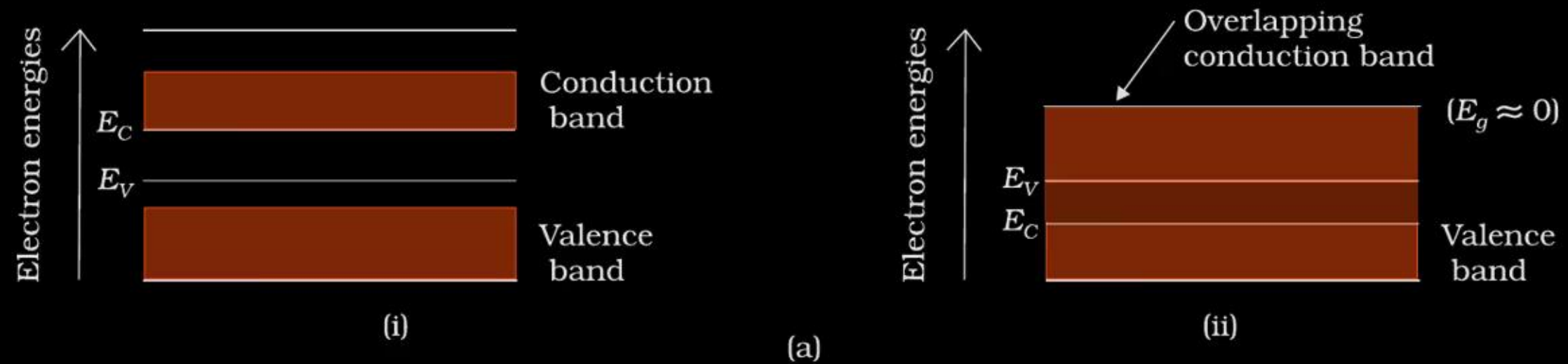
Classification on the Basis of Energy Bands

- **Metals:** CB and VB overlap or CB partially filled $\Rightarrow$ many free electrons

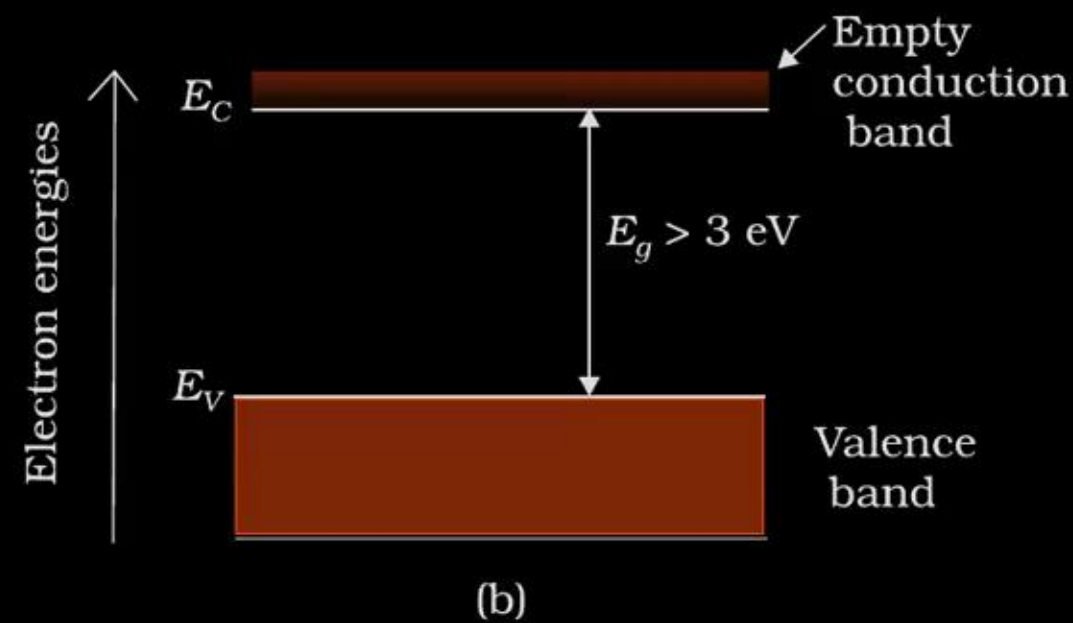


Classification on the Basis of Energy Bands

- **Metals:** CB and VB overlap or CB partially filled $\Rightarrow$ many free electrons

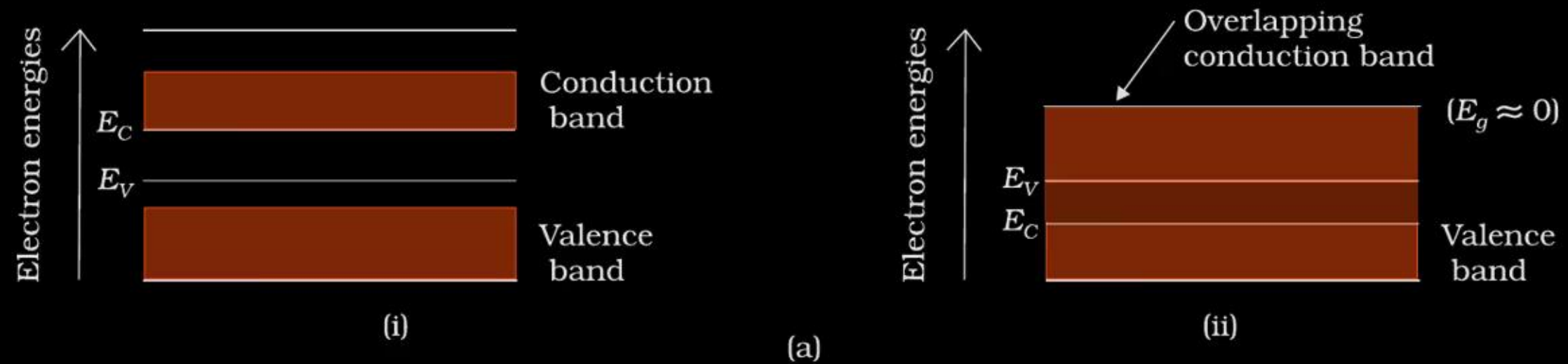


- **Insulators:** Large energy gap $E_g > 3 \text{ eV}$ between VB and CB $\Rightarrow$ no electron flow.



Classification on the Basis of Energy Bands

- **Metals:** CB and VB overlap or CB partially filled $\Rightarrow$ many free electrons



- **Insulators:** Large energy gap $E_g > 3 \text{ eV}$ between VB and CB $\Rightarrow$ no electron flow.
- **Semiconductors:** Small gap $E_g < 3 \text{ eV}$; some electrons can jump from VB to CB at room temperature.

